
Dimension, Not Solvability: Trained Matrix States Converge to the Minimal Faithful Representation Dimension

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Abstract

1 When do differently structured tasks drive a learner to the same representation?
2 We train identical matrix-state sequence models end-to-end on the word prob-
3 lems of five finite groups, S_3 , S_4 , A_5 , S_5 , A_6 , spanning the solvable/non-solvable
4 divide, and measure the effective rank of the learned state against each group’s
5 minimal faithful real representation dimension $d_{\min}$, the exact dimensionality rep-
6 resentation theory assigns the task. The learned rank converges onto the algebraic
7 minimum: group means 1.88/2.85/2.83/3.59/4.74 against $d_{\min} = 2/3/3/4/5$, all
8 19 seeds inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band, Spearman $\rho = 0.9747$, the
9 maximum this design permits under the S_4/A_5 tie. The designed head-to-head is
10 that tie: S_4 (solvable) and A_5 (non-solvable) share $d_{\min} = 3$, so if learned dimen-
11 sion were set by computational complexity class the pair should separate, and if
12 set by representation dimension it should coincide. A pre-registered Welch equiv-
13 alence test at ± 0.5 rank-units declares equivalence decisively (difference 0.019,
14 both one-sided tests near seven times the critical value). A pre-registered causal
15 force-rank razor then confirms the converged dimension is load-bearing rather
16 than a correlate of training: capping rank one below $d_{\min}$ zeroes exact recovery
17 in every group, and restoring $d_{\min}$ restores recovery. Learned representational
18 dimension follows the task’s algebra, not its complexity class.

19 1 Introduction

20 A central question for unifying accounts of neural representations is what, exactly, converges: when
21 systems trained on different problems arrive at similar internal structure, which property of the prob-
22 lems predicts the shared solution? We study a setting where the answer can be computed in closed
23 form before training. The word problem of a finite group G , mapping a sequence of generators to its
24 composed product, admits an exact algebraic description of the minimal structure any faithful solu-
25 tion must carry: $d_{\min}(G)$, the dimension of G ’s smallest faithful real linear representation. If trained
26 models converge to the task’s algebra, states trained on different groups should recruit representa-
27 tional dimension tracking $d_{\min}$; if instead the difficulty axis governs, dimension should sort with
28 the group’s computational class, solvable versus non-solvable, the divide at which word problems
29 become NC¹-complete [Barrington, 1989] and at which state-tracking expressivity results separate
30 recurrent architectures [Merrill et al., 2024, Grazi et al., 2025].

31 We train one architecture, a Transformer-encoded single-matrix-state model under a hard bottleneck,
32 on five groups spanning that divide, and measure the learned state’s restricted effective rank against
33 $d_{\min}$. Three results. First, convergence to the algebraic minimum: per-group mean rank lands within
34 4–8% of $d_{\min}$, all 19 seeds inside a pre-registered band, with Spearman $\rho = 0.9747$, the design’s
35 tie-capped maximum (Section 3). Second, the designed head-to-head at matched dimension: S_4

(solvable) and A_5 (non-solvable) share $d_{\min} = 3$, and a pre-registered equivalence test declares their learned ranks statistically equivalent, so the pair coincides on dimension rather than separating on complexity class (Section 4). Third, the converged dimension is causally load-bearing, not a correlate: capping rank at $d_{\min} - 1$ during training zeroes exact recovery in every group while $d_{\min}$ restores it (Section 5). A measurement lesson runs through all three: two instrument defects each produced a plausible wrong reading before being caught against raw run artifacts (Appendix B), and the convergence law above is what survived the repaired instrument.

2 Task, Model, and Instrument

Task. A word $w = g_{i_1} \cdots g_{i_L}$ is drawn as a random walk on G 's Cayley graph over a symmetric generating set ($L \sim U\{1, \dots, 8\}$) and must map to $\rho_G(g_{i_1} \cdots g_{i_L})$, the composed product's image under a pinned orthogonal reference representation of dimension $d_{\min}(G)$. The five groups are S_3, S_4, A_5, S_5, A_6 with $d_{\min} = 2, 3, 3, 4, 5$ (realizations in Appendix A); S_3 and S_4 are solvable, the rest are not.

Model. A Transformer encoder with d_{state} learned row-reader queries writes each word into a single state $Z(w) \in \mathbb{R}^{d_{\text{state}} \times d_{\text{state}}}$, with $d_{\text{state}} = d_{\min} + 2$ per group so that over-recruitment is expressible. The loss is cosine distance to the block-diagonal embedding of the reference ($\rho_G(\cdot) \oplus I_2$ in the observational sweep, $\rho_G(\cdot) \oplus 0$ in the causal wave); the readout is fixed, with no learned weights that could launder rank, and the decoder reads only Z , verified by a gradient blank-out test with a planted-leak negative control. Per-group step budgets (8k–40k) were pinned by pre-registered convergence bars before the decisional sweep ran.

Instrument. The learned state is defined only up to scale and an orthogonal change of basis (a Schur intertwiner): $Z(w) \approx cQ[\rho_G(w) \oplus \cdot]Q^\top$. We estimate the model's own dominant $d_{\min}$ -subspace U from the SVD of the *centered* covariance of $Z(w)$ over held-out words, measure **restricted effective rank**, the entropy effective rank of $U^\top Z(w)U$, and score **degauged recovery**, cosine to the reference after fitting (c, Q) on a fitting split and evaluating on a disjoint split, reported as $\text{rec}@0.9$ (the fraction of eval words above cosine 0.9). Rank is invariant under the fitted gauge, so a rank-deficient state cannot be degauged into a full-rank one; centering is load-bearing (Appendix B). The observational band $[0.7, 1.3] \cdot d_{\min}$, the equivalence margin and sample size, the force-rank grid and its recovery bar, and every falsifier were fixed before the decisive runs.

3 Convergence to the Algebraic Minimum

Figure 1 is the headline: unconstrained trained states recruit restricted effective rank 1.877 ± 0.060 , 2.852 ± 0.054 , 2.832 ± 0.062 , 3.591 ± 0.069 , 4.736 ± 0.023 (mean $\pm$ sd over seeds) against $d_{\min} = 2, 3, 3, 4, 5$, within 4–8% of the algebraic minimum at every group, with all 19 seeds inside the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band. Spearman correlation between learned rank and $d_{\min}$ is $\rho = 0.9747$, and this is the maximum the design can express: S_4 and A_5 tie at $d_{\min} = 3$ by construction, capping ρ below 1. Under the exact permutation null, $P(\rho \geq 0.8) = 8/120 \approx 6.7\%$. By pre-registration this observational leg is corroborating rather than independently decisive, because a $d_{\min}$ -restricted readout partially favors rank $\approx d_{\min}$ for near-orthogonal blocks; the causal weight is carried by Section 5. A disclosed length-robustness split moves per-group mean recovery cosine by at most 0.013 with no group-selective divergence (Appendix B).

4 Equivalence at Matched Dimension

The pair S_4/A_5 was designed into the family as the decisive contrast: the same $d_{\min} = 3$, opposite sides of the solvability divide, the axis on which expressivity theory separates state-tracking architectures [Merrill et al., 2024, Grazi et al., 2025, Siems et al., 2025]. If learned representational dimension were set by complexity class, the pair should separate; if by representation dimension, it should coincide. Because the interesting outcome is a null, the comparison was pre-registered as an equivalence test rather than a difference test: Welch two-one-sided tests on restricted effective rank at margin ± 0.5 rank-units (half the spacing of the $d_{\min}$ ladder), $n = 5$ seeds per group, with a pre-run power simulation showing 100% power to *reject* equivalence at a true gap of 1.0 rank-units, so a real class effect of one ladder step could not have hidden inside the margin.

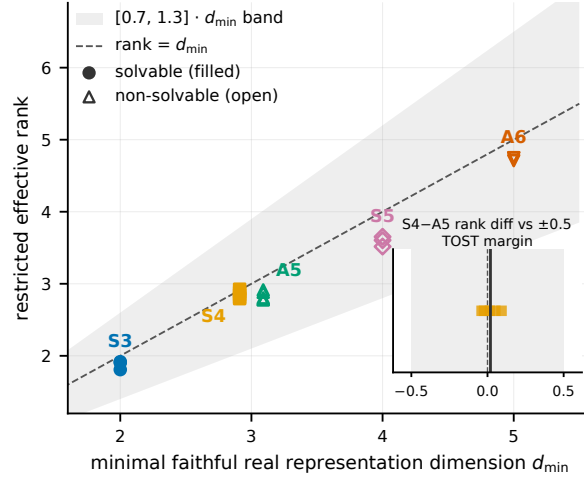


Figure 1: Learned rank converges to the minimal faithful representation dimension. Each point is one seed’s restricted effective rank on the group word problem vs. its group’s $d_{\min}$ (filled: solvable; open: non-solvable), with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band and the identity line; all 19 seeds are in band. Inset: per-seed $S_4 - A_5$ rank differences at matched $d_{\min} = 3$ against the pre-registered ± 0.5 rank-unit equivalence margin (shaded); the vertical line is the observed mean difference, $+0.019$.

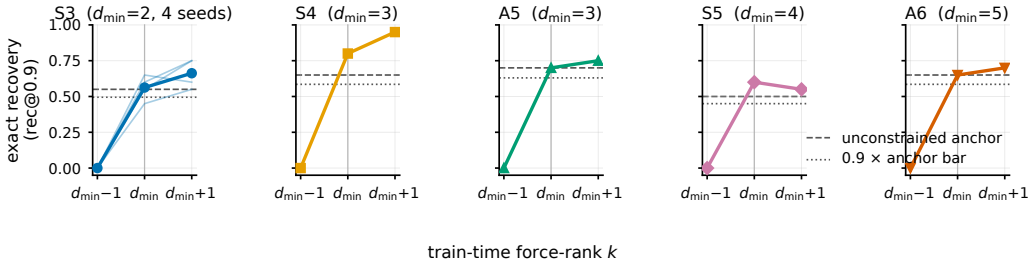


Figure 2: The converged dimension is causally load-bearing: per group, crosscheck $\text{rec}@0.9$ on held-out words at train-time force-rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$ (one seed per cell), vs. the unconstrained anchor (dashed) and the pre-registered $0.9 \times \text{anchor}$ bar (dotted). Recovery is exactly 0.000 at $d_{\min}-1$ in every group and returns at $d_{\min}$. S_3 overlays its four seeds (thin lines; bold mean); its below- $d_{\min}$ zero is unanimous.

86 The observed difference is $+0.019$ rank-units (se 0.037, df 7.8), and both one-sided tests pass at
87 roughly seven times the critical value ($t = 13.06$ and 14.12 against $t_{\text{crit}} = 1.865$): equivalence is
88 declared decisively. The two groups’ seed clusters are visually coincident in Figure 1 (inset). Con-
89 vergence here is governed by what the algebra demands, not by how hard the class is to compute.

90 5 The Causal Check

91 Convergence and equivalence are observational; the third leg intervenes. The encoder’s output is
92 spectrally truncated to rank $k \in \{d_{\min}-1, d_{\min}, d_{\min}+1\}$ throughout training against the zero-
93 padded target $\rho_G(\cdot) \oplus 0$ of rank exactly $d_{\min}$, alongside an unconstrained anchor from the same
94 family. Pre-registered reading: if $d_{\min}$ is load-bearing, $k = d_{\min}-1$ must fail and $k \geq d_{\min}$ must
95 recover past $0.9 \times$ the anchor’s $\text{rec}@0.9$ on the pre-pinned conservative readout; one below- $d_{\min}$ cell
96 recovering falsifies necessity.

97 Figure 2 shows a step function at $d_{\min}$. Necessity is exact and noiseless: $k = d_{\min}-1$ yields
98 $\text{rec}@0.9 = 0.000$ in all five groups and in all four independent seeds of the S_3 extension, while the

99 same cells’ recovery cosines (0.61–0.84) show the zero is a threshold, not a training collapse. Suf-
100 ficiency holds at $d_{\min}$: S_4 , A_5 , S_5 , A_6 recover past their bars outright (0.800/0.700/0.600/ 0.650
101 against 0.585/0.630/0.450/0.585), and S_3 , whose decisive cell landed inside a pre-stated ± 0.05
102 marginality trigger, was extended to four seeds under the trigger’s pre-registered routing: seed-
103 mean 0.5625 against the fixed 0.495 bar confirms the fifth group. An earlier 58-cell sweep of this
104 grid nulled through an instrument defect, an ambient identity block in the eye-padded target that
105 taxed every rank-capped arm; the verdict was registered as inconclusive with the tax established
106 from the raw artifacts, and the zero-padded wave above is the registered fix (Appendix B).

107 6 Related Work and Limitations

108 Merrill et al. [2024], Grazi et al. [2025], and Siems et al. [2025] frame state tracking by what
109 recurrent architectures *can* express on exactly these word problems, the solvability axis; this work
110 measures what trained states *do* converge to, and finds the learned dimension sorts by representation
111 theory instead. Nazari and Rusch [2026] and Sun et al. [2026] measure state-rank dynamics in
112 pretrained linear-attention models observationally, with no algebraic ground truth to converge to
113 and no causal leg. Nichani et al. [2025] prove associative-memory capacity under argmax decoding,
114 the loophole our exact-continuous-recovery scoring closes; Mishra et al. [2026] train matrix-state
115 RNNs on S_3 composition but measure no state rank and run no rank intervention.

116 **Limitations.** All results are on sub-1M-parameter synthetic models under a deliberate single-state
117 bottleneck; nothing here is a claim about pretrained networks, and the family is five groups with
118 $d_{\min} \in \{2, \dots, 5\}$ at $d_{\text{state}} = d_{\min} + 2$. Causal cells are single-seed except S_3 (four seeds), with
119 the necessity zeros exact wherever seeds exist; two groups at the shortest pinned budget show soft
120 convergence, disclosed in the run records, with the causal reading anchor-relative and unaffected.
121 The observational readout partially favors rank $\approx d_{\min}$ by construction, which is why it was pre-
122 registered as corroborating and the causal wave carries the claim.

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145 A The Five Groups and Reference Representations

G	$ G $	$d_{\min}$	solvable	realization of ρ_G
S_3	6	2	yes	standard (dihedral)
S_4	24	3	yes	cube rotations
A_5	60	3	no	icosahedral rotations
S_5	120	4	no	standard (zero-sum)
A_6	360	5	no	standard (zero-sum)

Table 1: The five task groups. $d_{\min}$ is the minimal faithful real representation dimension; each reference representation is orthogonal by construction and verified faithful (injective on G) at build time. S_4/A_5 form the designed pair: matched $d_{\min}$, opposite solvability.

146 B Instrument Details and the Two Defects

147 **Centering.** For an orthogonal target the uncentered covariance of $Z(w)$ is isotropic and carries no
 148 subspace information; the production lens centers it (a nontrivial irreducible representation has zero
 149 group mean, cancelling the constant block). The step is load-bearing: an uncentered lens scores a
 150 flawless synthetic model 0.705 where the centered lens scores 0.9996 on identical data.

151 **Length-robustness split.** The disclosed split referenced in Section 3 scores only words of length
 152 $L \geq 2$ (the $L = 1$ read is attention-degenerate and was demoted with a mechanism note in the run
 153 records): it moves per-group mean recovery cosine by at most 0.013 (per-seed at most 0.041), with
 154 no group-selective divergence.

155 **The ambient-identity tax.** The first 58-cell sweep’s force-rank target was the eye-padded $\rho_G(\cdot) \oplus I_2$,
 156 whose rank is $d_{\text{state}} = d_{\min} + 2$ for every group. The constant identity block is the cheapest reliable
 157 loss reduction, so a rank-capped arm buys it first, leaving effective budget $k - 2 < d_{\min}$ for ρ_G at
 158 every grid point; no capped cell in that sweep ever tested the confirm direction. The raw-artifact
 159 signature: the 39 force-rank cells’ direct cosine matched the rank- k optimum $\sqrt{k/d_{\text{state}}}$ with mean
 160 absolute deviation 0.028 (maximum 0.166; the two S_5 below- $d_{\min}$ cells are the only outliers, a
 161 disclosed optimization shortfall), so the arms had trained to their rank-constrained ceiling rather
 162 than under-training. The sweep verdict was registered as inconclusive with the tax as mechanism;
 163 the zero-padded causal wave of Section 5 is the registered fix, its 30 cells configuration-verified
 164 one-by-one against a manifest re-derived independently from the design record. In that wave an
 165 eye-padded corroboration arm reproduces the failure on demand at raw $k = d_{\min} + 1$ (effective
 166 budget $d_{\min} - 1$) while the tax-paid point recovers.

167 C Per-Seed Observational Rank

G	$d_{\min}$	n	restricted eff. rank	band	in band
S_3	2	3	1.877 ± 0.060	[1.4, 2.6]	3/3
S_4	3	5	2.852 ± 0.054	[2.1, 3.9]	5/5
A_5	3	5	2.832 ± 0.062	[2.1, 3.9]	5/5
S_5	4	3	3.591 ± 0.069	[2.8, 5.2]	3/3
A_6	5	3	4.736 ± 0.023	[3.5, 6.5]	3/3

Table 2: Restricted effective rank of the unconstrained trained state (mean $\pm$ sd over seeds) against each group’s $d_{\min}$, with the pre-registered $[0.7, 1.3] \cdot d_{\min}$ band. Every seed of every group is in band; Spearman $\rho = 0.9747$ is the tie-capped maximum; the dimension-matched pair S_4/A_5 differs by 0.019 rank-units.